Inverted MAML Card 1: Round 20 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 30%/40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±90 LSB

1. Round 20 Summary & Cumulative Comparison (R1–R1)

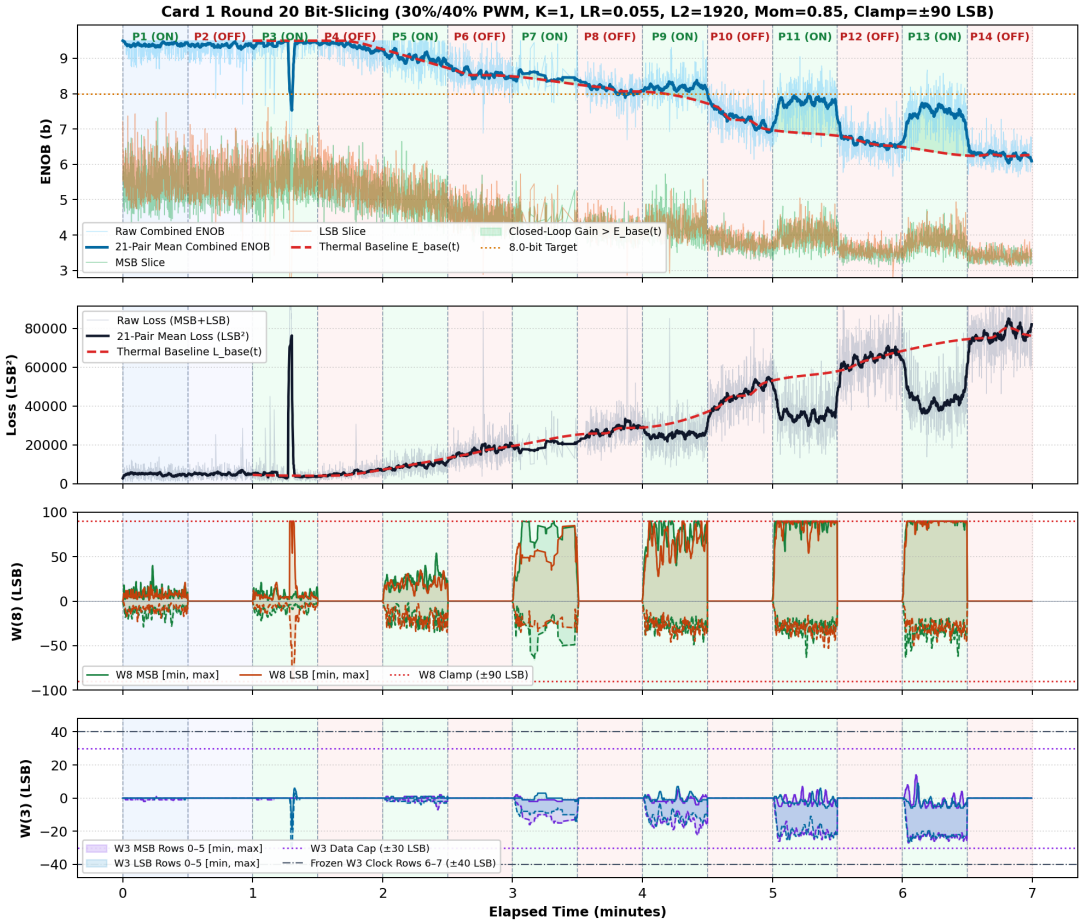
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 20 Detrended Trend Metrics:** Lift vs E_base = **+0.48 b**, Loss Ratio = **1.42x**, Climb = **+0.63 b**, Jitter = **0.36 b**, Recovery = **0.89**, Score = **+1.691**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1*	0.055	1920	0.85	±90	+0.48b	1.42x	+0.63b	0.36b	0.89	+1.691

2. Phase-by-Phase Telemetry (4,629 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	461	9.48->9.41 (9.39b)	-	4138->4443	[40,23]	2	460
P2	Cold / Regler OFF	406	9.36->9.37 (9.37b)	-	4976->5177	[0,0]	0	0
P3	30% Heat / Regler ON	389	9.34->9.45 (9.33b)	-0.05 b	5136->3683	[22,90]	30	389
P4	30% Heat / Regler OFF	388	9.37->9.24 (9.34b)	-0.10 b	3936->6575	[0,0]	0	0
P5	30% Heat / Regler ON	377	9.19->8.97 (9.02b)	+0.15 b	6776->10196	[54,35]	7	377
P6	30% Heat / Regler OFF	355	8.74->8.51 (8.57b)	+0.00 b	12863->18585	[0,0]	0	0
P7	30% Heat / Regler ON	115	8.40->8.41 (8.49b)	+0.07 b	17908->21164	[90,85]	16	115
P8	30% Heat / Regler OFF	305	8.27->8.02 (8.11b)	-0.05 b	24460->29841	[0,0]	0	0
P9	40% Heat / Regler ON	327	8.15->8.20 (8.19b)	+0.38 b	27268->25478	[90,90]	22	327
P10	40% Heat / Regler OFF	308	7.76->7.09 (7.31b)	+0.01 b	36286->51047	[0,0]	0	0
P11	40% Heat / Regler ON	311	7.18->7.78 (7.74b)	+0.94 b	48193->35056	[90,90]	26	311
P12	40% Heat / Regler OFF	299	6.83->6.53 (6.63b)	+0.01 b	58998->67082	[0,0]	0	0
P13	40% Heat / Regler ON	300	6.80->7.41 (7.43b)	+1.14 b	59739->43652	[90,90]	27	300
P14	40% Heat / Regler OFF	288	6.31->6.22 (6.27b)	-0.02 b	74882->78536	[0,0]	0	0


Card 1 Round 20 Bit-Slicing (30%/40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±90 LSB)
P1 (ON)
P2 (OFF)
P3 (ON)
P4 (OFF)
P5 (ON)
P6 (OFF)
P7 (ON)
P8 (OFF)
P9 (ON)
P10 (OFF)
P11 (ON)
P12 (OFF)
P13 (ON)
P14 (OFF)
ENOB (b)
Raw Combined ENOB
21-Pair Mean Combined ENOB
MSB Slice
LSB Slice
Thermal Baseline E_base(t)
Closed-Loop Gain > E_base(t)
8.0-bit Target
Loss (LSB²)
Raw Loss (MSB+LSB)
21-Pair Mean Loss (LSB²)
Thermal Baseline L_base(t)
W(8) (LSB)
W8 MSB [min, max]
W8 LSB [min, max]
W8 Clamp (±90 LSB)
W(3) (LSB)
W3 MSB Rows 0–5 [min, max]
W3 LSB Rows 0–5 [min, max]
W3 Data Cap (±30 LSB)
Frozen W3 Clock Rows 6–7 (±40 LSB)
Elapsed Time (minutes)

3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 20

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±90 LSB, while W(3) rows 0–5 stay within ±30 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 20 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

